

DNA Sequence Assembly Report

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1. Overview

This project implements a robust DNA sequence assembly pipeline in Python (`project.py`) designed to process, validate, model, and reconstruct primary DNA sequence structures from raw Sanger sequencing data. The application operates by parsing raw information from standardized CSV source files (`DNA_x_k.csv`), executing multi-tiered data cleaning rules, serializing data matrices into dictionary-mapped JSON structures, and building directed de Bruijn graphs from sequence k-mers.

Using topological metrics, the pipeline validates graph structural connectivity and uses a stack-based implementation of Hierholzer's algorithm to resolve an Eulerian path. It then successfully assembles the ultimate target DNA strand string and generates both a visual graph network layout (`DNA_x.png`) and a text report containing the reconstructed sequence result (`DNA_x.txt`).

The software system relies on a stable, production-grade open-source scientific stack: pandas manages high-throughput matrix manipulations and cleaning, NetworkX (MultiDiGraph) provides underlying data structures for handling overlapping multi-edge graphs, matplotlib exports visual topology layouts, and a comprehensive suite of unit test modules is implemented with pytest (`project_test.py`).

Core Architectural Pipeline:

- **Data Ingestion & Cleaning:** `read_csv` → `clean_data`
- **Graph Synthesis & Rendering:** `generate_sequences` → `construct_graph` → `plot_graph`
- **Sequence Reconstruction & Export:** `is_valid_graph` → `construct_dna_sequence` → `save_output`

2. Implementation Details

- **read_csv:** Ingests raw Sanger sequencing records using `pandas.read_csv`. It enforces a strict, zero-header schematic mapping the structured data columns exactly to: `['SegmentNr', 'Position', 'A', 'C', 'G', 'T']`. This vectorized data load sets up highly optimized matrix workflows for downstream formatting operations.
- **clean_data:** Applies an iterative multi-pass filter across individual sequence elements to isolate noise from valid genomic segments:
 - Nucleotide Flag Validation: Ensures high-fidelity data quality by filtering

out rows failing to contain exactly one hot-encoded base value (where the row sum across columns A, C, G, and T strictly equals 1).

- **Coordinate Continuity Verification:** Detects and flags gaps by confirming that structural segment positions create an unbroken sequential integer range tracking precisely from 1 up to $\max(\text{Position})$.
- **Duplicate & Conflict Resolution:** Cross-checks matching sequence indices. Identical values are safely collapsed into a single row using `drop_duplicates`, while conflicting structural data readings at identical coordinates result in the immediate rejection of the entire corrupted segment.
- **Redundant Sequence Purging:** Builds a local lookup cache using an internal helper routine (`_segment_to_sequence`) to translate fragments into literal text strings, discarding redundant segment matching patterns.
- **generate_sequences:** Extracts the remaining filtered rows, transforming data groups into an intermediate key-value lookup structure indexed directly by stringified segment identifiers (`segment_id` $\rightarrow$ `sequence_string`). This mapping is then serialized into an exportable JSON configuration via `json.dumps`.
- **construct_graph:** De-serializes the JSON strings and extracts overlapping k-mer window slices of length k. It uses a directed `nx.MultiDiGraph` layout where each moving window slice generates a directed edge leading directly from its left-hand (k-1)-mer prefix to its right-hand (k-1)-mer suffix. Implementing a multi-graph framework is vital to preserve recurring edge paths across repetitive genomic configurations.
- **plot_graph:** Uses `networkx.spring_layout` to compute node coordinates, drawing structural topologies through `matplotlib` before releasing memory with `close` commands. This provides clean visual spacing for evaluating network characteristics.
- **is_valid_graph:** Performs edge-case structural sanity checks. It confirms the network is not empty, checks for weak connectivity across nodes via `nx.is_weakly_connected`, and inspects node degrees to verify Eulerian properties. The graph is marked valid if and only if the count of nodes with mismatched incoming and outgoing degrees is exactly 0 (an Eulerian cycle) or exactly 2 (an Eulerian path with clearly defined start/end points).
- **construct_dna_sequence:** Performs path reconstruction using a stack-based Hierholzer's algorithm on a duplicate graph instance. For configurations containing unbalanced endpoints, a temporary structural edge bridges the sink back to the source to resolve a complete cycle. The algorithm traverses the graph via a memory-efficient stack, sequentially removing edges to build the path array. Once the path is reversed and any duplicate endpoints are trimmed, it constructs the final DNA string by taking the full string of the first node and appending the final character of each subsequent node.
- **save_output:** A basic I/O file-writer that commits the processed genomic

string or error messages directly to text files on disk.

- **main:** Acts as the primary CLI orchestration entrypoint. It reads input file paths from `sys.argv`, uses regular expressions to extract metrics (x index and k window bounds) from filenames, runs the pipeline functions sequentially, and writes out the file results.

3. Evaluation

- **Implemented Specifications:** Tasks 1 through 8 are fully operational and accessible via standard terminal invocations: `python project.py csv-files/DNA_x_k.csv`.
- **System Correctness:** Reconstructs sequences properly when evaluating sample data. Valid graph topologies produce accurate, full-length target strings, while invalid structural layouts (such as disconnected networks) are safely caught and output a standardized error warning.
- **Omissions:** Bonus Tasks 9 (`construct_all_dna`), 10 (`align_segments`), and 11 (`generate_all_alignments`) were excluded from this core implementation layout.
- **Operational Limitations:** String matches assume explicit, high-fidelity overlaps. The algorithm does not currently feature alignment scoring configurations for partial edge match states or single-base mismatches.

4. Example Runs

4.1 DNA_1_5.csv

Terminal Execution Command:

```
python project.py csv-files/DNA_1_5.csv
```

Pipeline Execution Phase	Operational Outcome Metric
<code>read_csv</code>	72 raw data rows populated into active DataFrame memory
<code>clean_data</code>	Pruned outlier coordinates; preserved valid rows across segments 1–8
<code>generate_sequences</code>	Formatted sequence translation output mapping dictionary strings to JSON
<code>construct_graph</code>	Instantiated de Bruijn graph configuration (k=5); 26 nodes, 40 edges mapped
<code>plot_graph</code>	Generated and exported system topology layout visualization directly to <code>DNA_1.png</code>
<code>is_valid_graph</code>	Evaluated as True (Valid weakly connected layout; matches boundary balance limits)

Pipeline Execution Phase	Operational Outcome Metric
construct_dna_sequence	Reconstructs Eulerian path: TTAA -> TAAT -> AATT -> ATTA -> ... -> ACTG
save_output	Committed 44-character DNA string to text file DNA_1.txt

Reconstructed Primary DNA Output Sequence String (DNA_1.txt):

TTAATTACTCACTGGGTCCTGGCTAATTACTCACTACG-
CACTG

Network Topology Visualization (DNA_1.png)

Below is the structural layout map generated by the pipeline's plotting routines, displaying the interconnected node transitions and overlapping paths resolved during execution:

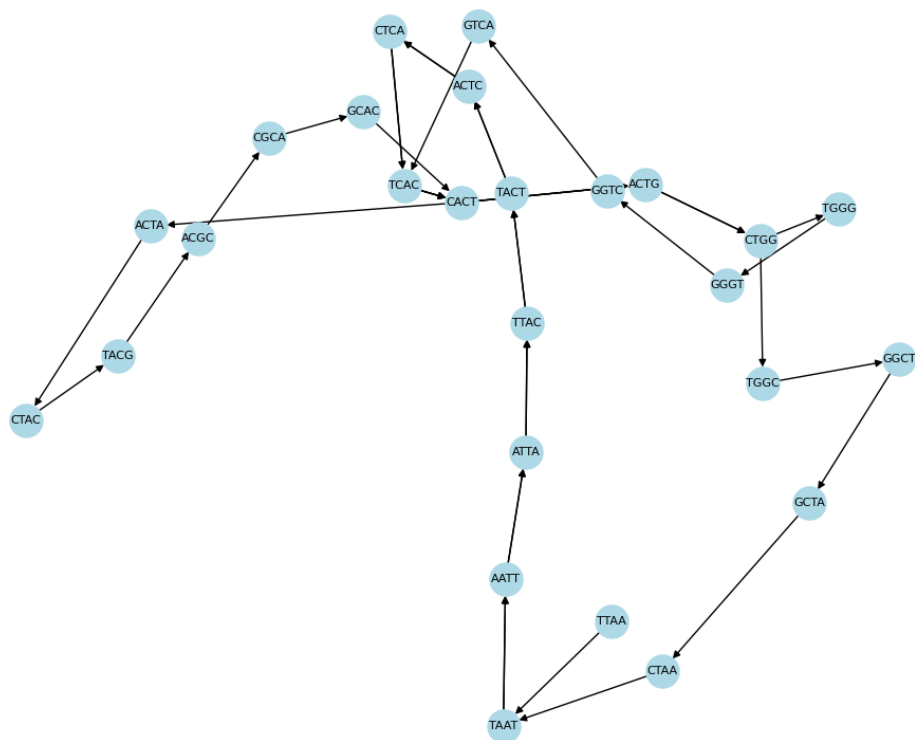


Figure 1: Generated de Bruijn graph topology structure layout for DNA_1_5.csv (26 nodes, 40 edges).

4.2 DNA_5_5.csv (Disconnected Input/Error Path Handling)

Terminal Execution Command:

```
python project.py csv-files/DNA_5_5.csv
```

Pipeline Execution Phase	Operational Outcome Metric
read_csv	Ingested target data entries (12 total data rows)
clean_data	Retained segments 1 and 2 following structural validity filters
generate_sequences	Generated JSON layout string: {"1": "ATGAA", "2": "CTGAATG"}
construct_graph	Evaluated 5 internal nodes alongside 4 data path links (k=5)
is_valid_graph	Evaluated as False (Aborts processing due to weak connectivity failure)
construct_dna_sequence	Skips path routing; returns structured sequence building error exception
save_output	Logs failure notice text to DNA_5.txt: "DNA sequence can not be constructed."

5. Challenges

- **Multi-Edge Disambiguation:** Resolving graphs with repeated structural patterns required strict edge management during graph traversals. Reusing identical node transitions can stall path tracking or cause infinite loops if edges are not removed as they are traversed.
- **Vectorized Data Profiling Hierarchy:** Ensuring all four programmatic cleaning operations executed in the correct order without dropping valid data segments required a careful tracking design using conditional filters before dropping rows.
- **Dynamic Stream Traversal Operations:** Modifying graph topologies within construct_dna_sequence required using graph duplicate copies (graph.copy()) to prevent mutating original network graphs needed for rendering or secondary analysis.

6. Tests Description

The validation ecosystem uses pytest configuration matrices to isolate components and verify performance across clear boundary test limits.

Target Test Identifier	Parameter Test Scope Matrix	Verification Goal
<code>test_clean_data</code>	Evaluates sequences with incorrect nucleotide rows, missing position offsets, or conflicting values.	Confirms robust parsing and filtering under noisy data conditions.
<code>test_generate_sequences</code>	Evaluates single segment array inputs, multi-segment vectors, and varying length character arrays.	Confirms precise formatting of exported JSON data strings.
<code>test_construct_graph</code>	Tests k-mer slicing transformations using varying lookback windows (k=3, k=5).	Validates edge generation accuracy inside de Bruijn networks.
<code>test_is_valid_graph</code>	Evaluates complete Eulerian cycles, open Eulerian paths, disconnected components, and invalid degree balances.	Ensures improper topologies are caught before sequence reconstruction.
<code>test_construct_dna_sequence</code>	Tests path reconstructions across diverse structural patterns, looping cycles, and short pathways.	Confirms the accuracy of the stack-based Hierholzer's algorithm.

Validation Suite Metric Summary: Executing `pytest project_test.py -v` confirms complete coverage across all parameterized conditions, showing zero regressions across the codebase (16 tests passed).

Verification and Packaging Steps

1. **Local Test Execution:** Run the full validation suite to verify code stability via `pytest project_test.py -v`.
2. **Archive Verification:** Ensure your final submission zip file maintains a flat root structure, bundling `project.py`, `project_test.py`, `report.docx`, and your source data files without subdirectories.